Carbon-Aware AutoML Experiment Report

1. Executive Summary & Provenance Legend

Generated: 2026-11-11T19:29:41.035202+00:00 Recommended Model: decision_tree | Primary Score [MEASURED]: 1.0 Legend: [MEASURED] = timing/metrics | [ESTIMATED] = CPU energy | [CALCULATED] = CO2e & stats

2. Dataset Fingerprint & Integrity Verification

Source: D:\MYSELF\GRP\Carbon-Emission\datasets\child_health_westbengal_1000.csv SHA-256 [MEASURED]: c68a778e13cc257f80f9edb8c44ea2b42eb9c30fcbbcbb9cce3d4c4bbb1d7ca8 Shape [MEASURED]: (1000, 8) | Original DataFrame Unchanged: True

3. Preprocessing & Validation Methodology

Strategy: stratified_kfold (folds=5, repeats=3, seed=42) Excluded Columns: [] Preprocessing fit strictly on training folds/split (zero test leakage).

Dataset Profile & Leakage Analysis

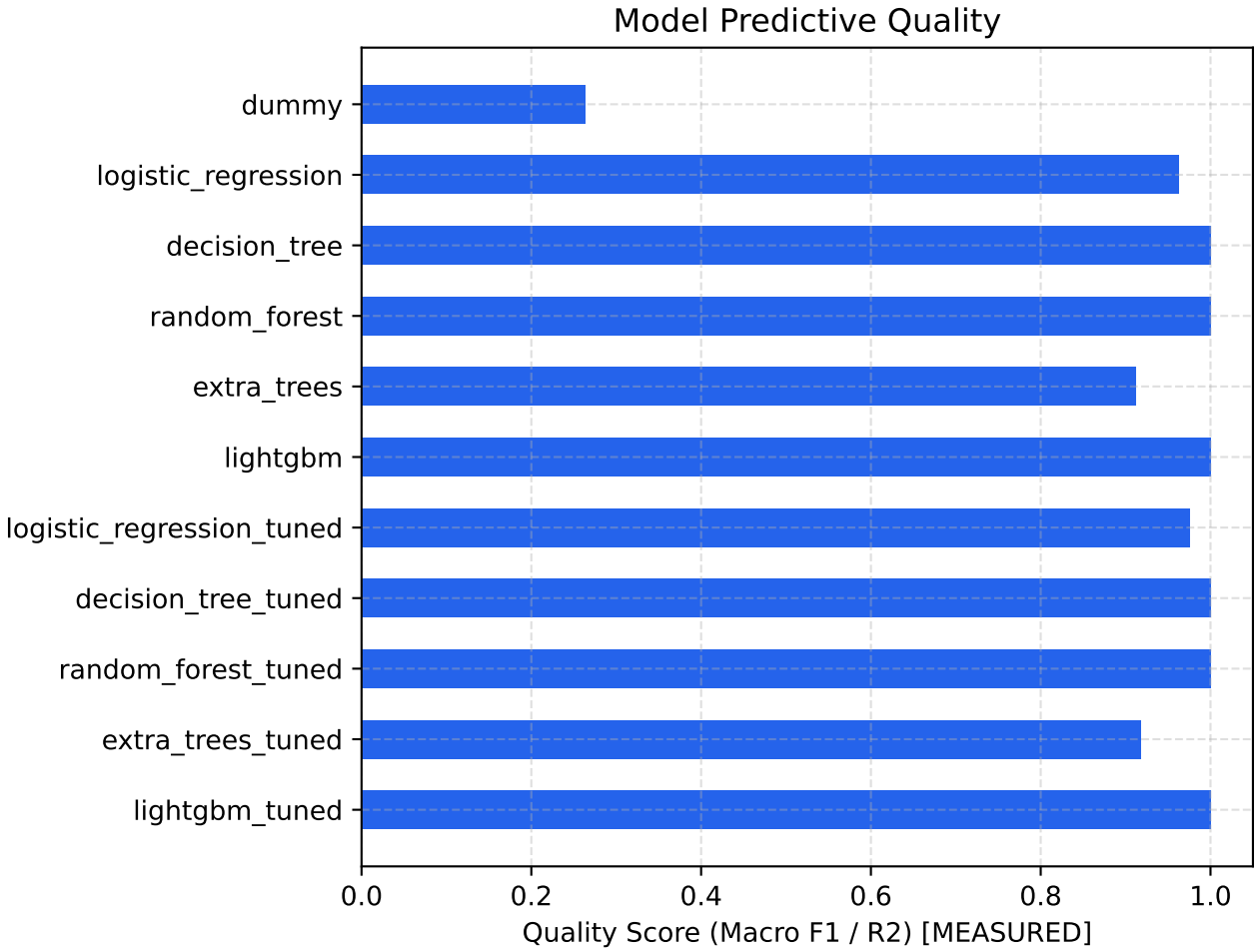
4. Dataset Profile Summary [MEASURED]

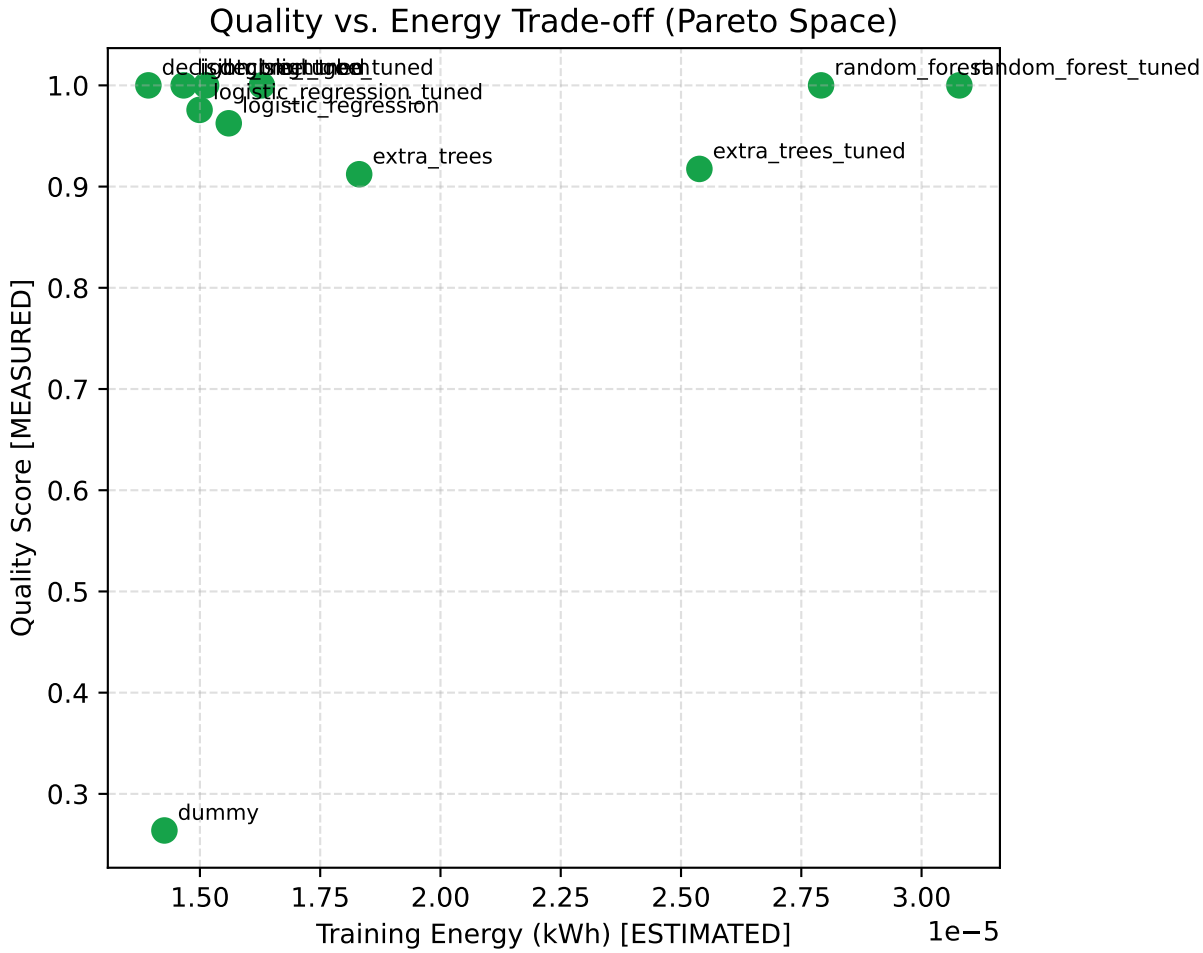
Task suggestion: classification | Memory: 0.2147 MB | Duplicates: 0 - Age (months): numerical (missing=0) - Gender: binary (missing=0) - Height (cm): numerical (missing=0) - Weight (kg): numerical (missing=0) - Subdivision (West Bengal): categorical (missing=0) - Mother's Education: categorical (missing=0) - Mid-Upper Arm Circumference (cm): numerical (missing=0)

5. Leakage Warnings

[CRITICAL] Mid-Upper Arm Circumference (cm): single_feature_performance — Mid-Upper Arm Circumference (cm) achieves 1.0000 accuracy alone using DecisionTree(max_depth=3) (DummyClassifier baseline: 0.6550). [HIGH] Mid-Upper Arm Circumference (cm): mutual_information_anomaly — Feature 'Mid-Upper Arm Circumference (cm)' accounts for 97.8% of total mutual information across features (MI=0.8767 of total 0.8962). [CRITICAL] Mid-Upper Arm Circumference (cm): perfect_range_separation — Feature 'Mid-Upper Arm Circumference (cm)' has completely non-overlapping value ranges per class (Severe: [8.4, 11.4], Moderate: [11.5, 12.4], Normal: [12.5, 19.1]).

Model Comparison & Pareto Frontier





Energy, Carbon & Statistical Analysis

6. Training Energy by Phase

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  ingestion: 0.008980800164863467s | 1.0813445975738593e-05 kWh [MEASURED]   preprocessing:
0.020461100153625 01s | 1.026217208103387e-05 kWh [MEASURED]   hpo_logistic_regression:
0.7339982998091727s | 1.7050331489300644e-05 kWh [MEASURED]   hpo_decision_tree:
0.1141152000054717s | 1.1621714785678254e-05 kWh [MEASURED]   hpo_random_forest:
8.164004700025544s | 0.00010165587134370371 kWh [MEASURED]   hpo_extra_trees:
3.408215400064364s | 4.6297160222531634e-05 kWh [MEASURED]   hpo_lightgbm:
1.4409410001244396s | 2.394985987057204e-05 kWh [MEASURED]   final_training:
0.0021885999012738466s | 1.1692525952551743e-05 kWh [MEASURED]
```

7. CO2e Estimates [CALCULATED] & Carbon Intensity Assumption

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Total Energy [ESTIMATED]: 0.0008811830574253319 kWh Total CO2e [CALCULATED]:
0.0006168281401977323 kg Carbon Intensity: 0.7 kg CO2e/kWh (source: user_configured)
Boundary: Direct CPU/RAM compute only (excludes PUE, cooling, network, manufacturing).
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8. Inference Energy [ESTIMATED]

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Per 1k predictions (kWh): N/A Per prediction (J): 0.03464784045807995 | Uncertainty: ±100.0%
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9. Carbon Budget & Statistical Tests [CALCULATED]

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Budget limit (kg): None Pairwise Wilcoxon tests recorded: 0
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Model Recommendation, Carbon Passport & Limitations

10. Final Recommendation

Recommended Model: decision_tree Justification: 'decision_tree' achieves the best normalised balance of macro_f1 (1.0000) and training_energy_kwh (0.000014 kWh) among Pareto-optimal models that pass the quality gate (minimum macro_f1 = 0.995). Quality Violations: ['dummy', 'logistic_regression', 'extra_trees', 'logistic_regression_tuned', 'extra_trees_tuned'] Pareto Optimal: ['decision_tree']

11. Carbon Passport Summary

Model ID: f9e394f1-0bcd-46d2-b978-702a2061764f | Version: carbon_automl 1.0.0 Dataset Hash: c68a778e13cc257f80f9edb8c44ea2b42eb9c30fcbbcbb9cce3d4c4bbb1d7ca8

12. Known Limitations & Threats to Validity

• Grid carbon intensity (0.7 kg CO2e/kWh, user_configured) is static. • Sub-second CPU energy measurements carry ±50–100% uncertainty from background noise. • Critical target leakage warning(s) were active during this run.